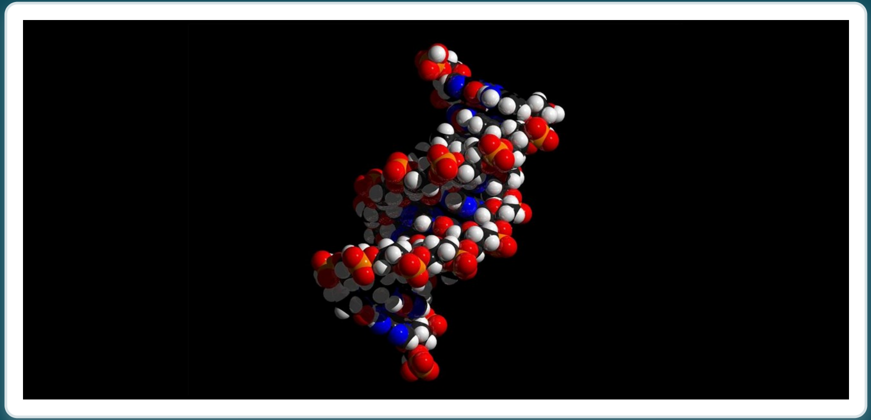
CHAPTER 2

The Chemical Level of Organization



Atoms, bonds, reactions, and the molecules of life.

Anatomy & Physiology · Broward-Miami Health Institute

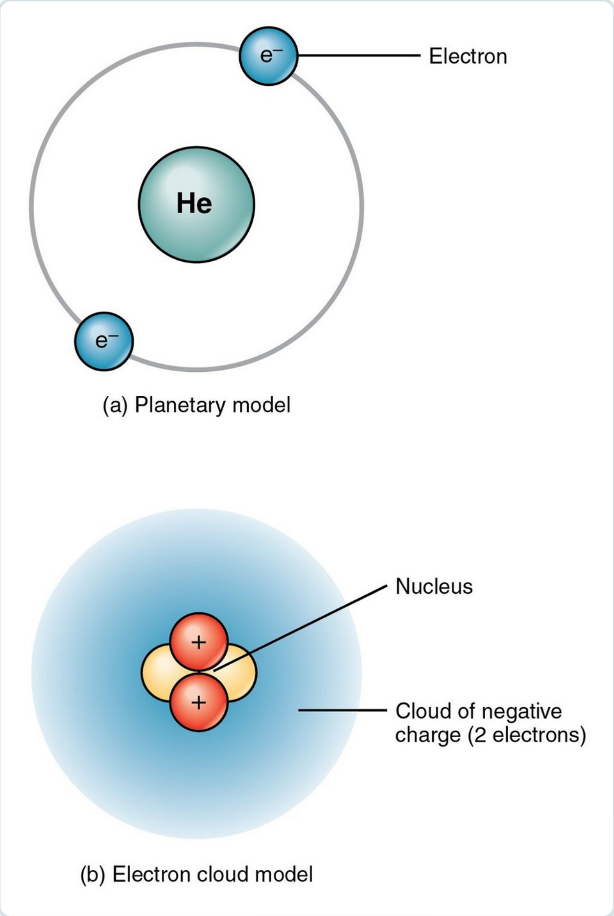
FACULTY EDITION

CHAPTER 2

Learning Objectives

After studying this chapter, students will be able to:

1. Describe the fundamental composition of matter
2. Identify the three subatomic particles
3. Identify the four most abundant elements in the body
4. Explain the relationship between an atom's number of electrons and its relative stability
5. Distinguish between ionic bonds, covalent bonds, and hydrogen bonds
6. Explain how energy is invested, stored, and released via chemical reactions, particularly those reactions that are critical to life



Atoms — the building blocks of all matter.

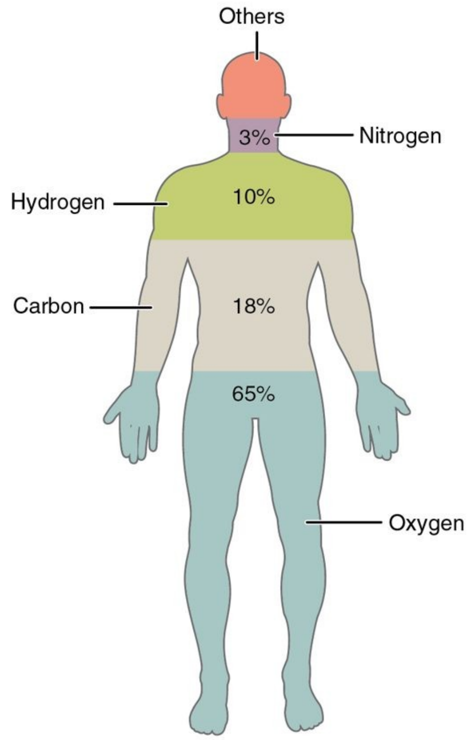
2.1

SECTION

Elements and Atoms: The Building Blocks of Matter

What everything is made of — elements, atoms, and isotopes.

Elements in the Human Body

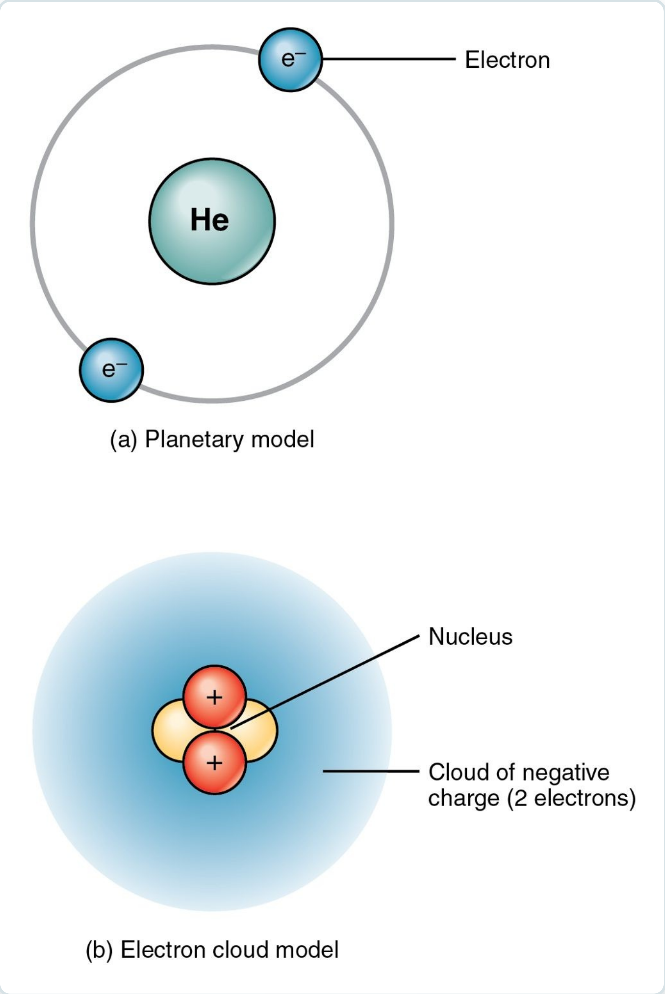


Element	Symbol	Percentage in Body
Oxygen	O	65.0
Carbon	C	18.5
Hydrogen	H	9.5
Nitrogen	N	3.2
Calcium	Ca	1.5
Phosphorus	P	1.0
Potassium	K	0.4
Sulfur	S	0.3
Sodium	Na	0.2
Chlorine	Cl	0.2
Magnesium	Mg	0.1
Trace elements include boron (B), chromium (Cr), cobalt (Co), copper (Cu), fluorine (F), iodine (I), iron (Fe), manganese (Mn), molybdenum (Mo), selenium (Se), silicon (Si), tin (Sn), vanadium (V), and zinc (Zn).		less than 1.0

The major and trace elements of the body

- Four elements make up ~96% of body mass
- Oxygen, carbon, hydrogen, nitrogen
- Trace elements are vital in tiny amounts
- Matter has mass and takes up space

The Structure of an Atom



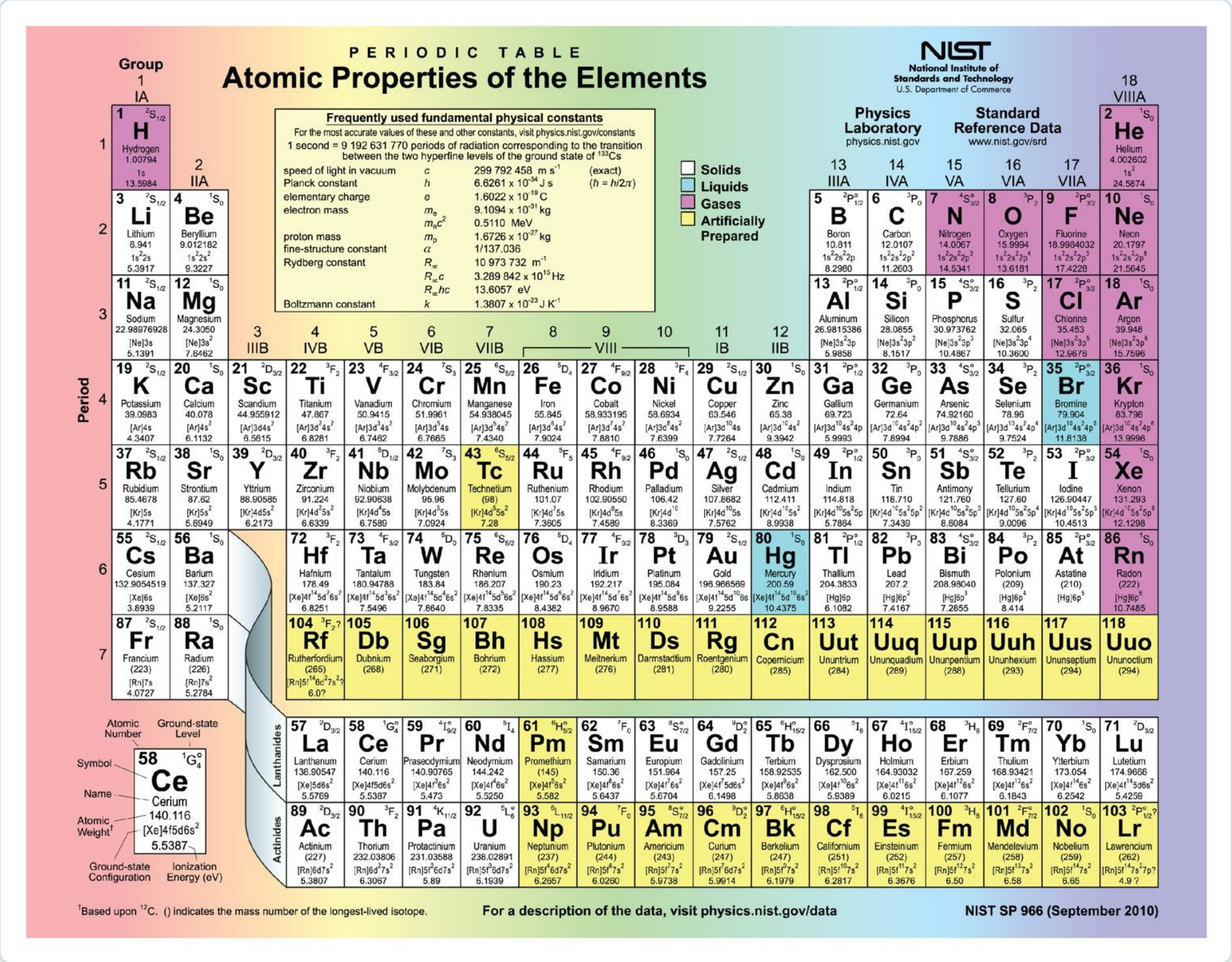
Planetary and electron-cloud models of the atom

- Protons, neutrons, and electrons
- Nucleus holds protons (+) and neutrons
- Electrons (−) occupy outer shells
- Atomic number = number of protons

SECTION 2.1

The Periodic Table

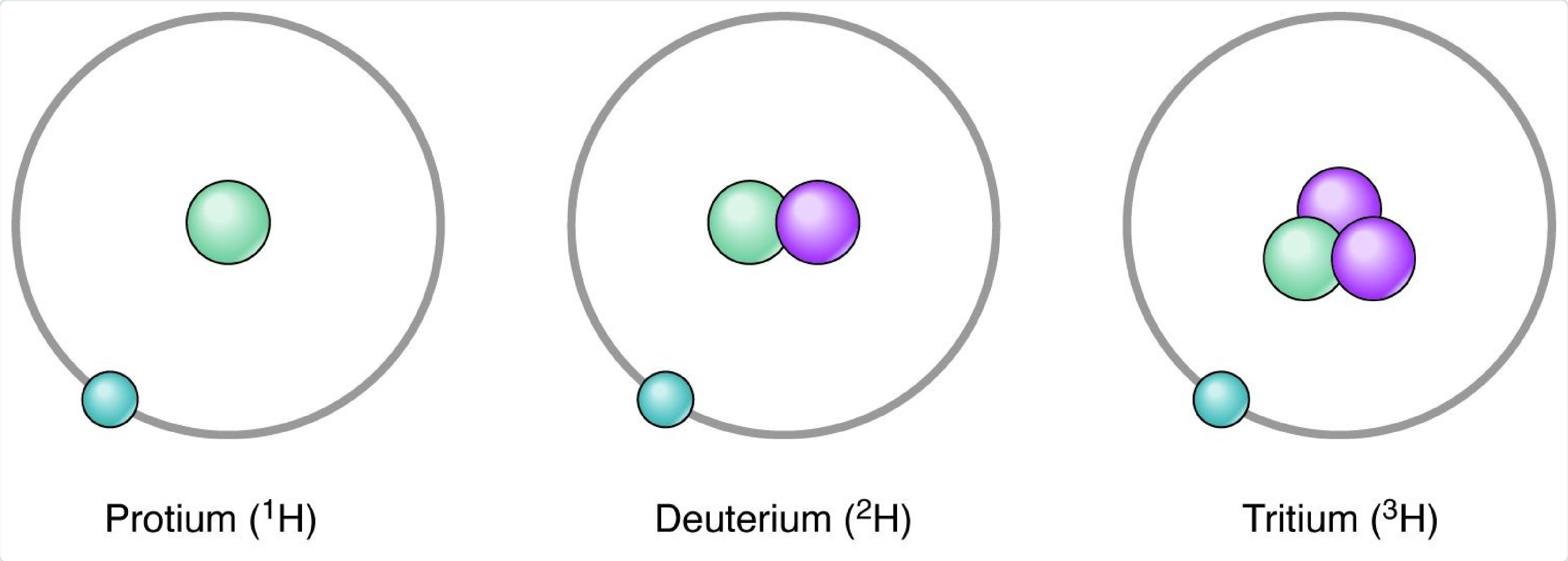
- Organizes all known elements
- Arranged by atomic number
- Groups share chemical behavior
- About two dozen elements occur in the body



The periodic table of the elements

SECTION 2.1

Isotopes



Isotopes of hydrogen (protium, deuterium, tritium)

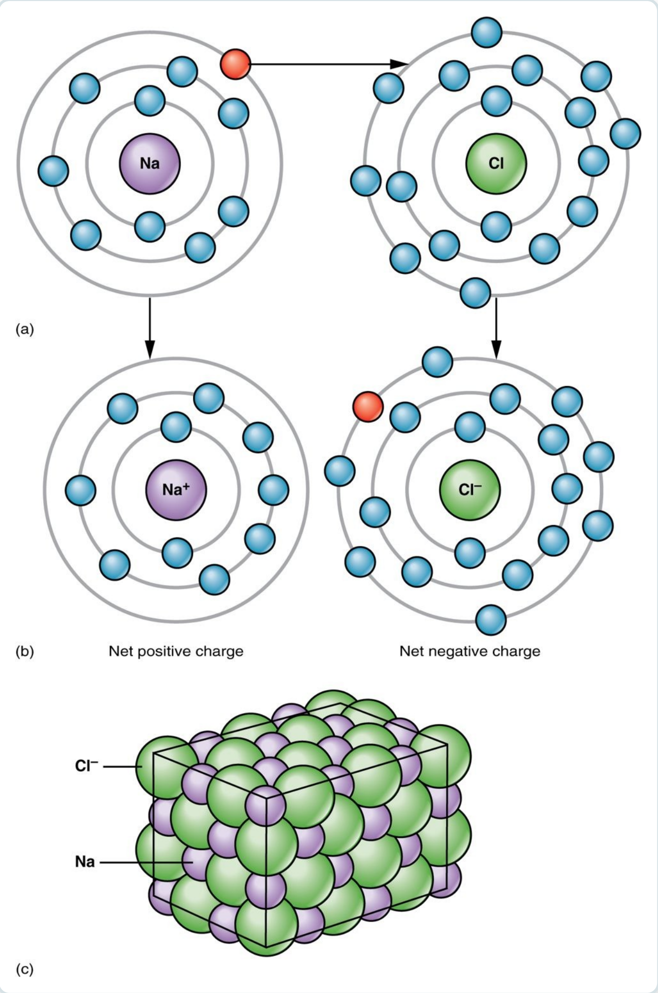
- Same element, different neutron count
- Some isotopes are radioactive
- Radioisotopes used in imaging and therapy
- Example: the isotopes of hydrogen

SECTION

2.2 Chemical Bonds

How atoms join together — ionic, covalent, and hydrogen bonds.

Ionic Bonds



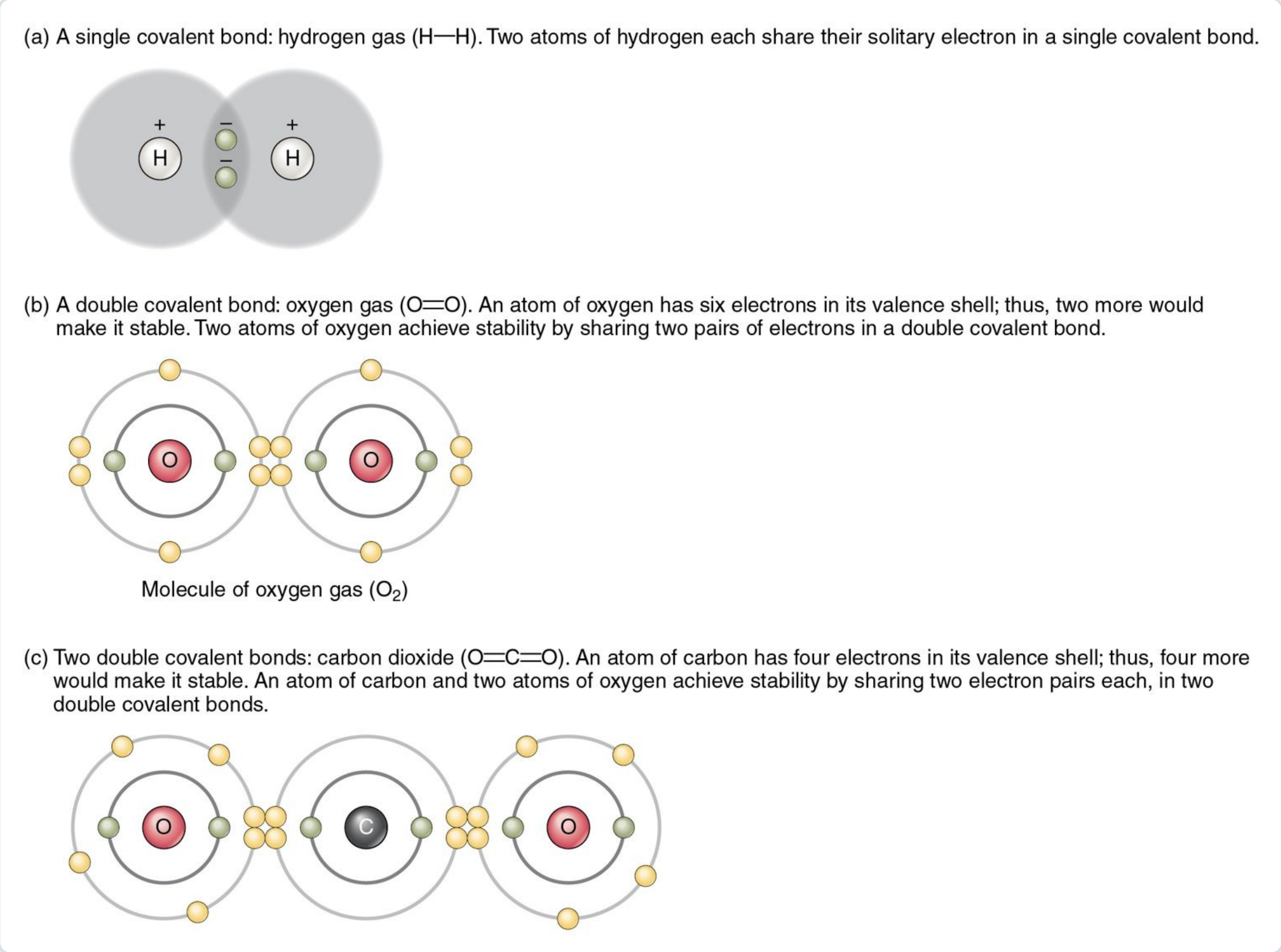
Ion formation and ionic bonding

- Electrons are transferred between atoms
- Creates charged ions (cations and anions)
- Opposite charges attract
- Example: Na^+ and Cl^- form salt

SECTION 2.2

Covalent Bonds

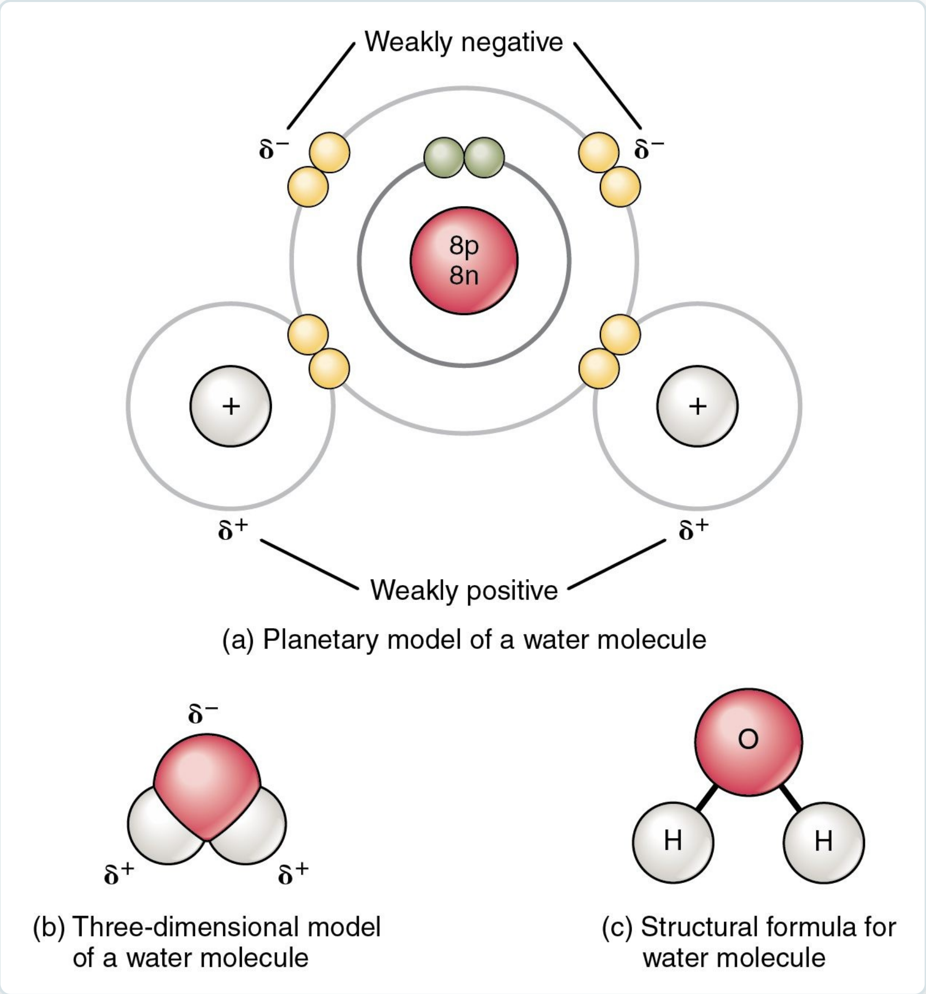
- Atoms share pairs of electrons
- Single, double, or triple bonds
- Strong and stable
- The backbone of organic molecules



Single and double covalent bonds

SECTION 2.2

Polar Bonds & Water



Polarity of water and hydrogen bonding

- Unequal sharing makes water polar
- Slightly positive and negative regions
- Hydrogen bonds link water molecules
- Source of water's unique properties

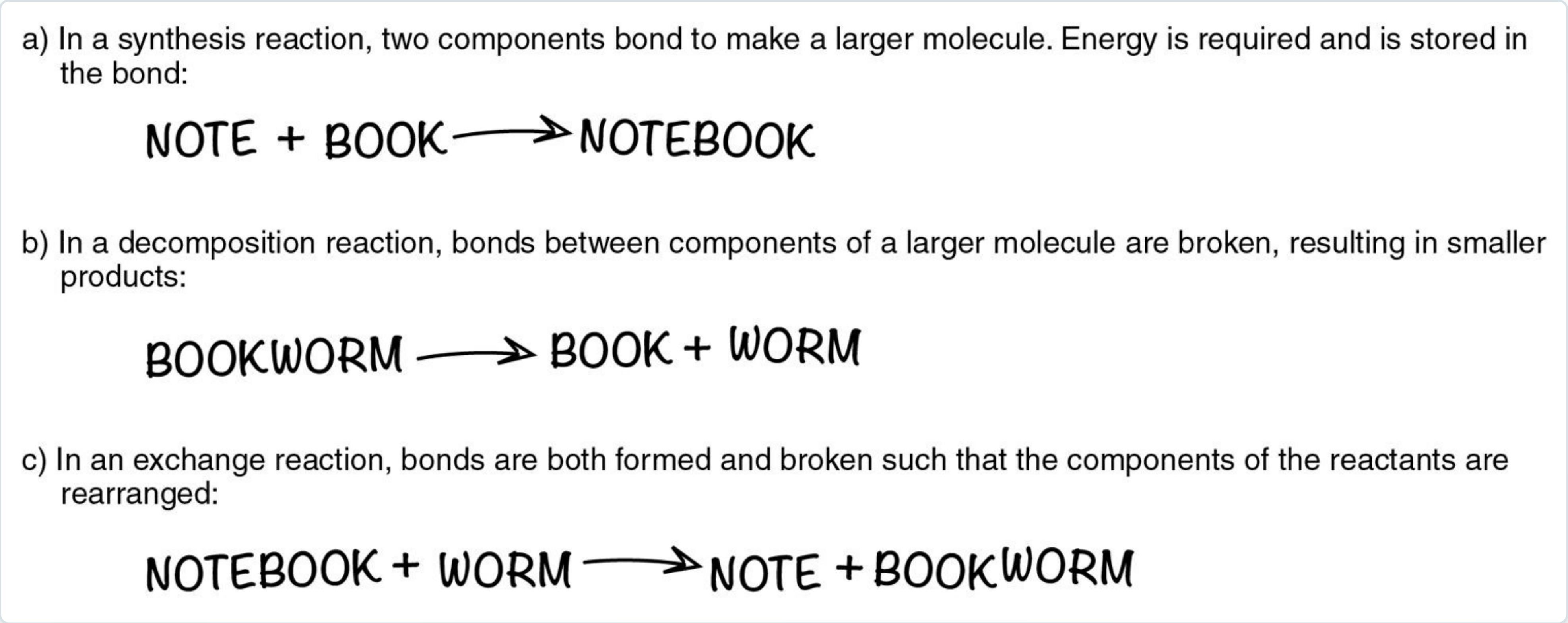
2.3

SECTION

Chemical Reactions

How matter is rearranged — synthesis, decomposition, and the role of enzymes.

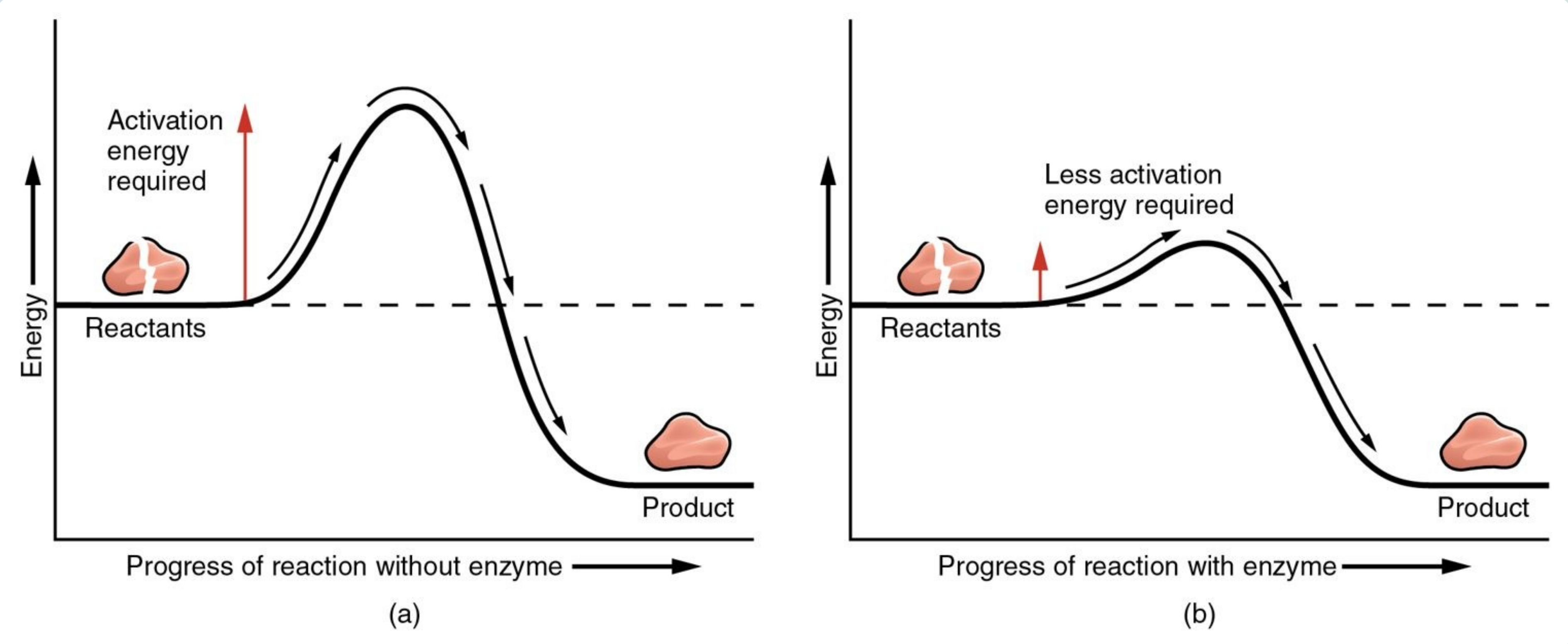
Types of Chemical Reactions



Synthesis, decomposition, and exchange reactions

- Synthesis builds larger molecules
- Decomposition breaks molecules apart
- Exchange reactions swap components
- Some reactions are reversible

Enzymes & Activation Energy



Enzymes lower the activation energy of reactions

- Reactions need activation energy to start
- Enzymes lower that energy barrier
- Speed reactions without being used up
- Most are proteins ending in -ase

2.4

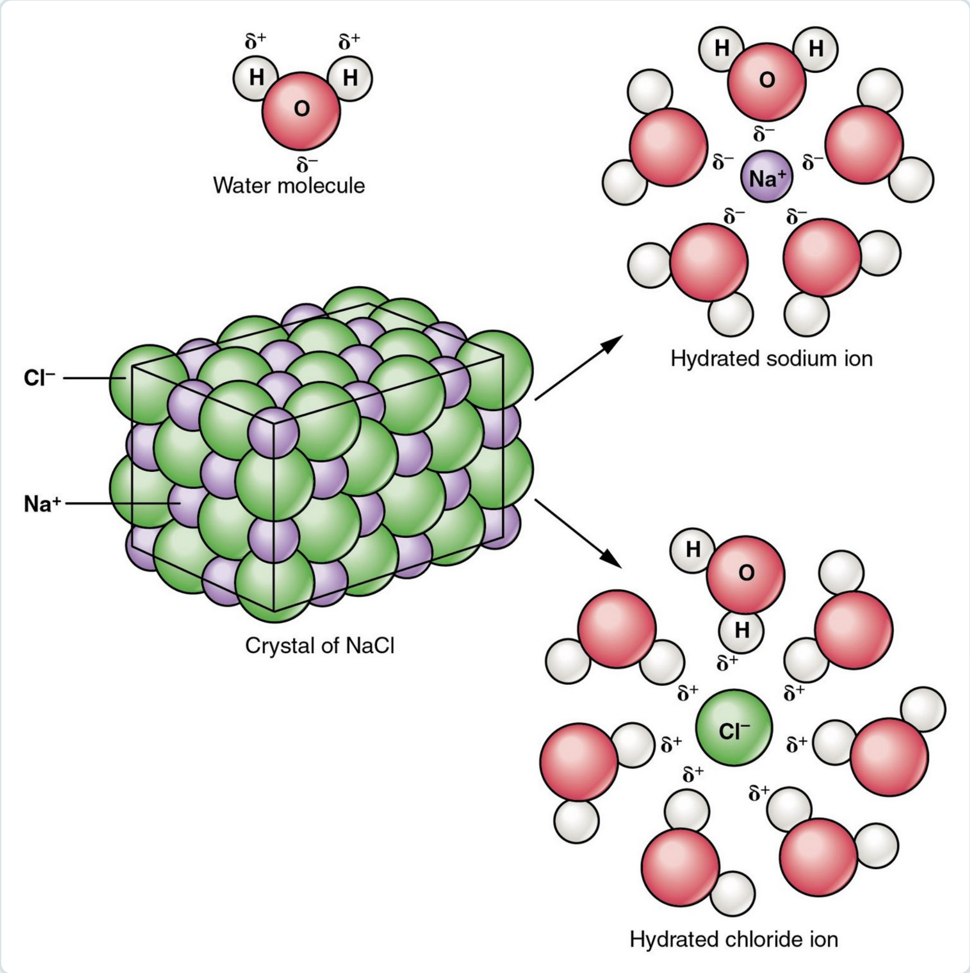
SECTION

Inorganic Compounds Essential to Human Functioning

Water, acids, bases, and the pH the body must keep in balance.

Water as the Body's Solvent

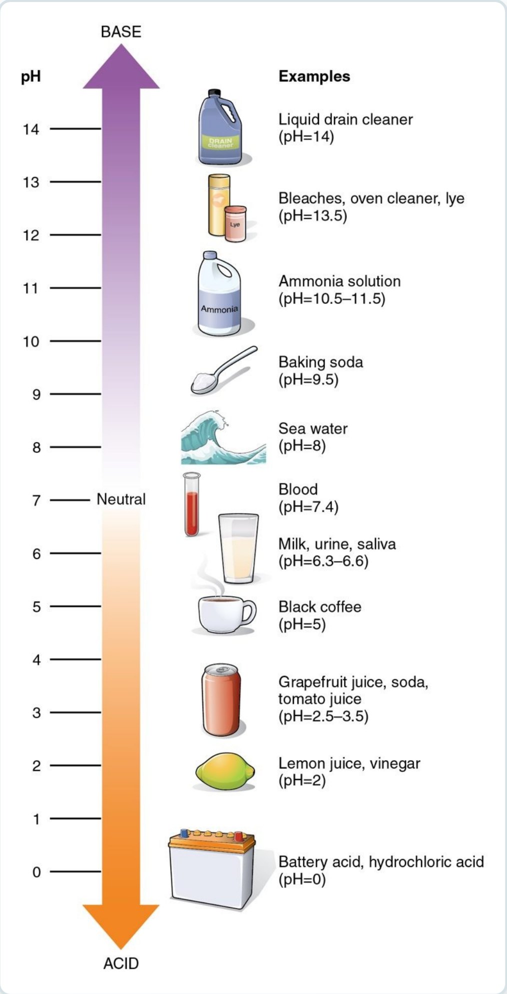
- Water is the universal biological solvent
- Polar substances dissolve readily
- Ions are surrounded (hydrated)
- The medium for nearly all reactions



Sodium chloride dissolving in water

SECTION 2.4

The pH Scale



The pH scale with common substances

- Measures acidity and alkalinity (0–14)
- Lower is acidic; higher is basic
- Blood pH held near 7.4
- Buffers resist pH change

SECTION

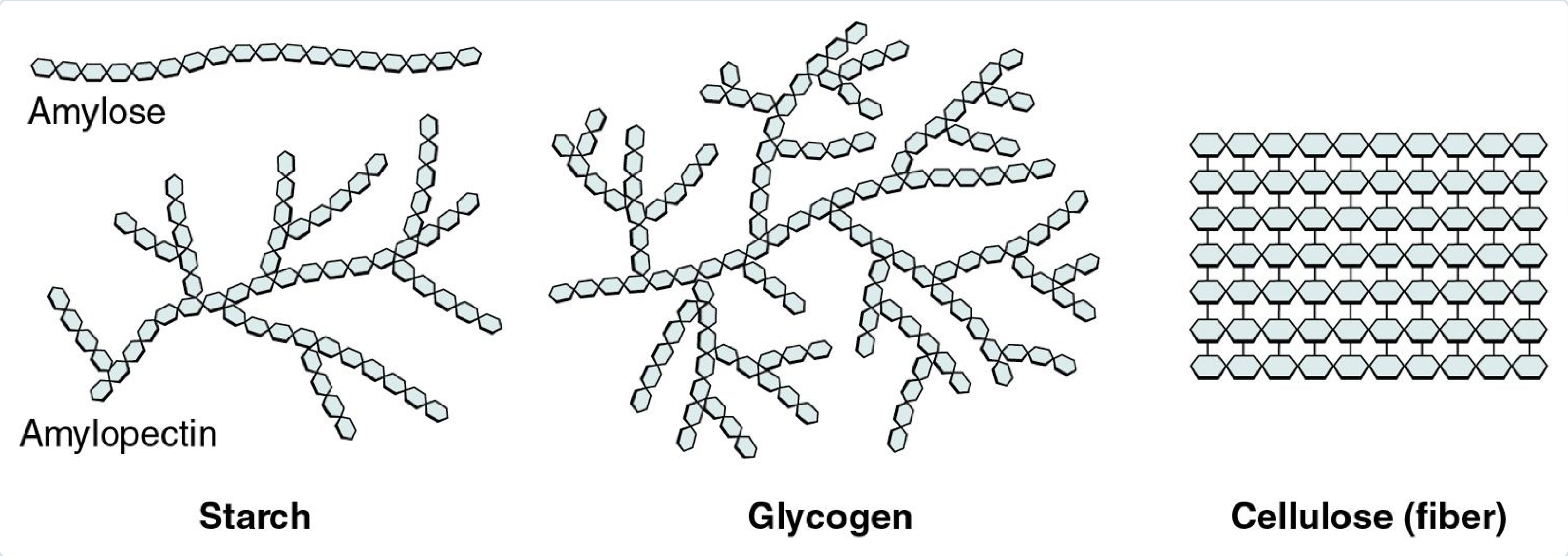
2.5

Organic Compounds Essential to Human Functioning

The four macromolecules of life — carbohydrates, lipids, proteins, and nucleic acids.

SECTION 2.5

Carbohydrates

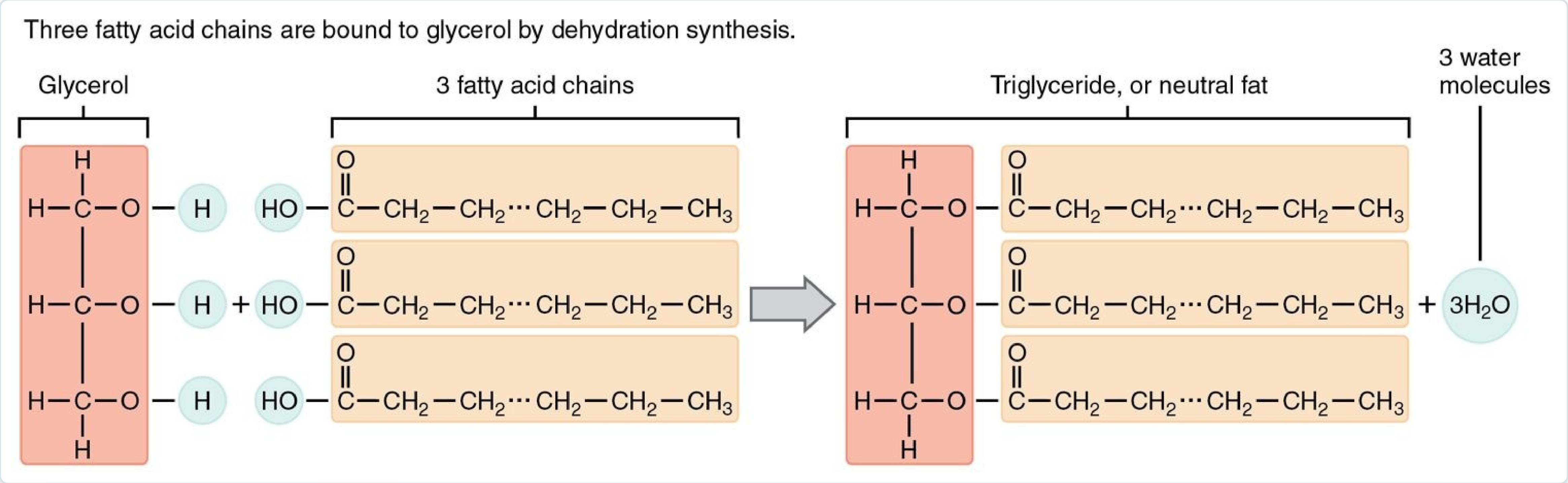


Polysaccharides: starch, glycogen, and cellulose

- Made of carbon, hydrogen, and oxygen
- Monosaccharides build polysaccharides
- Glucose is the body's main fuel
- Stored as glycogen

SECTION 2.5

Lipids

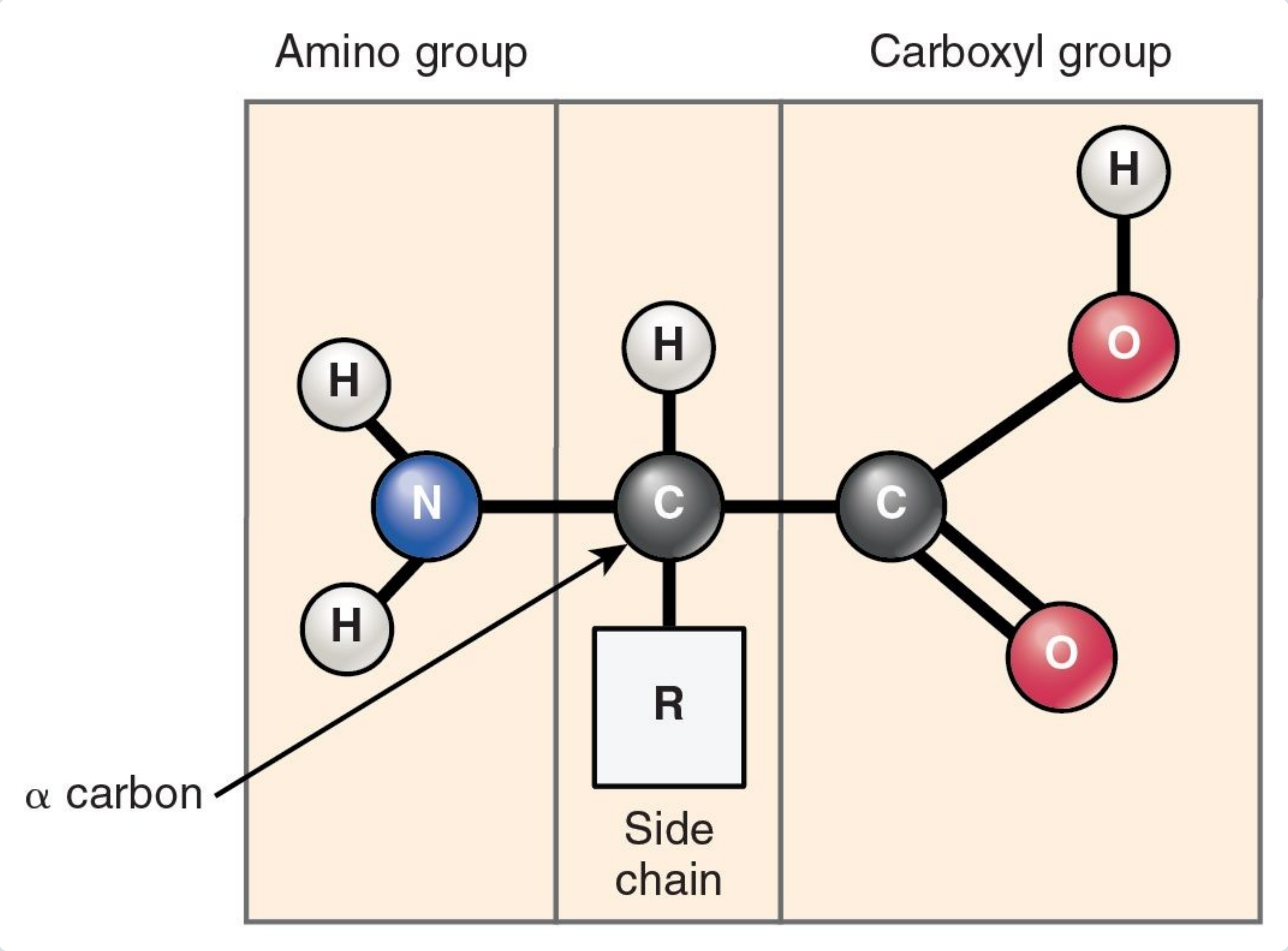


Saturated and unsaturated fatty acids

- Hydrophobic — do not dissolve in water
- Triglycerides store energy
- Saturated vs. unsaturated fatty acids
- Phospholipids build cell membranes

Proteins — Amino Acids

- Built from 20 amino acids
- Amino group + carboxyl group + R side chain
- Joined by peptide bonds
- The side chain gives each its identity

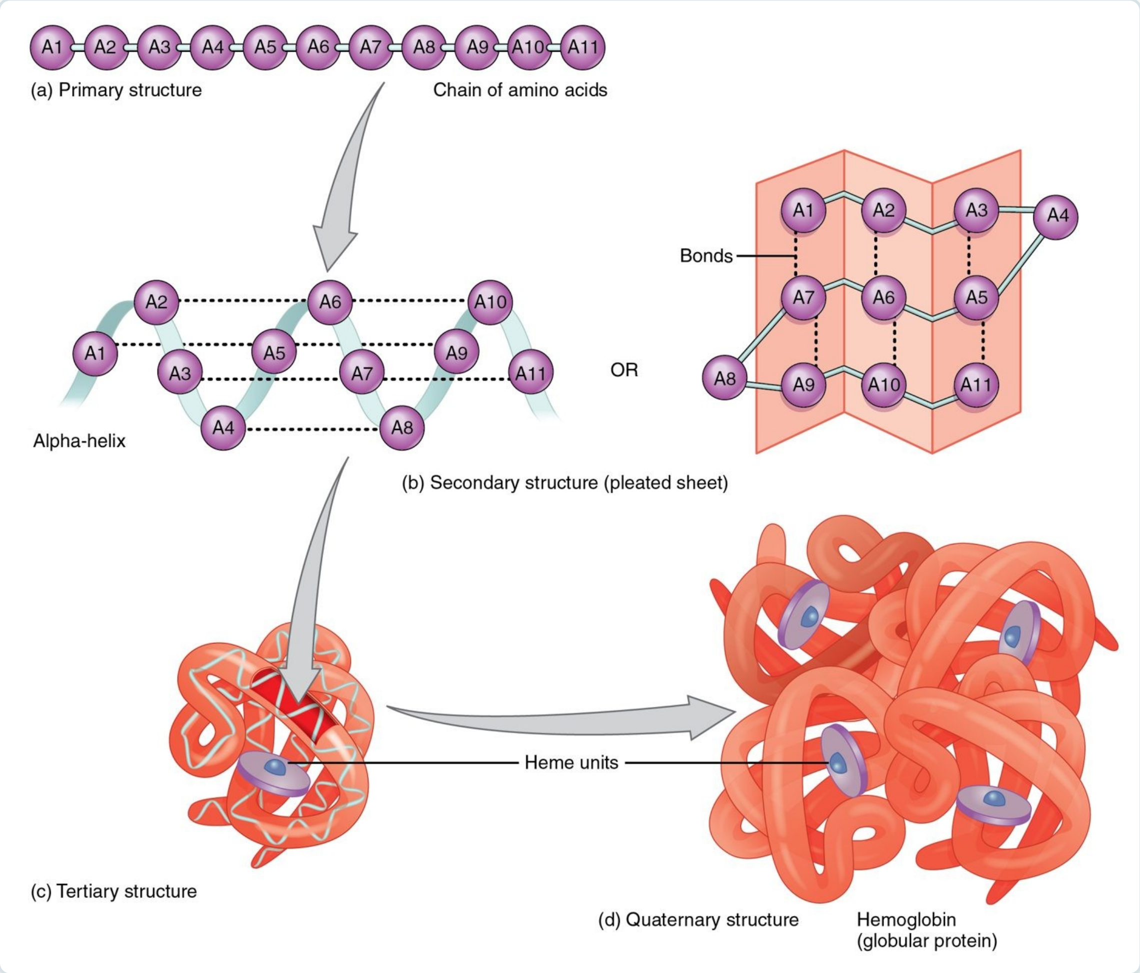


The general structure of an amino acid

SECTION 2.5

Protein Structure & Function

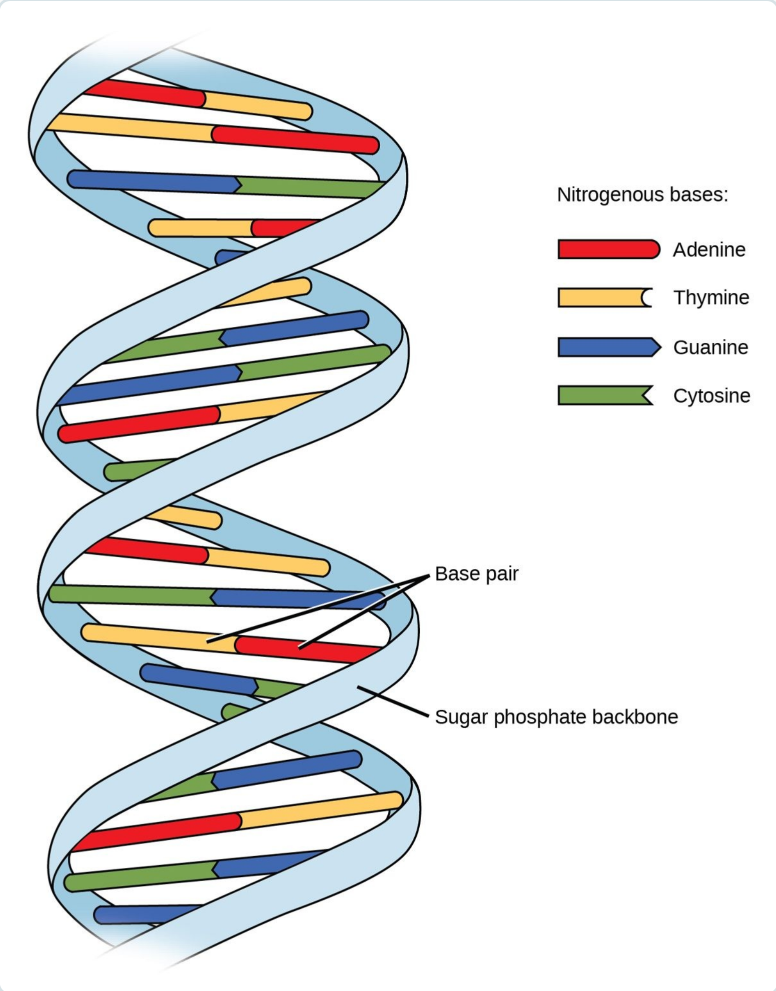
- Four levels: primary to quaternary
- Shape determines function
- Heat or pH can denature proteins
- Enzymes, antibodies, and more



The four levels of protein structure

SECTION 2.5

Nucleic Acids — DNA



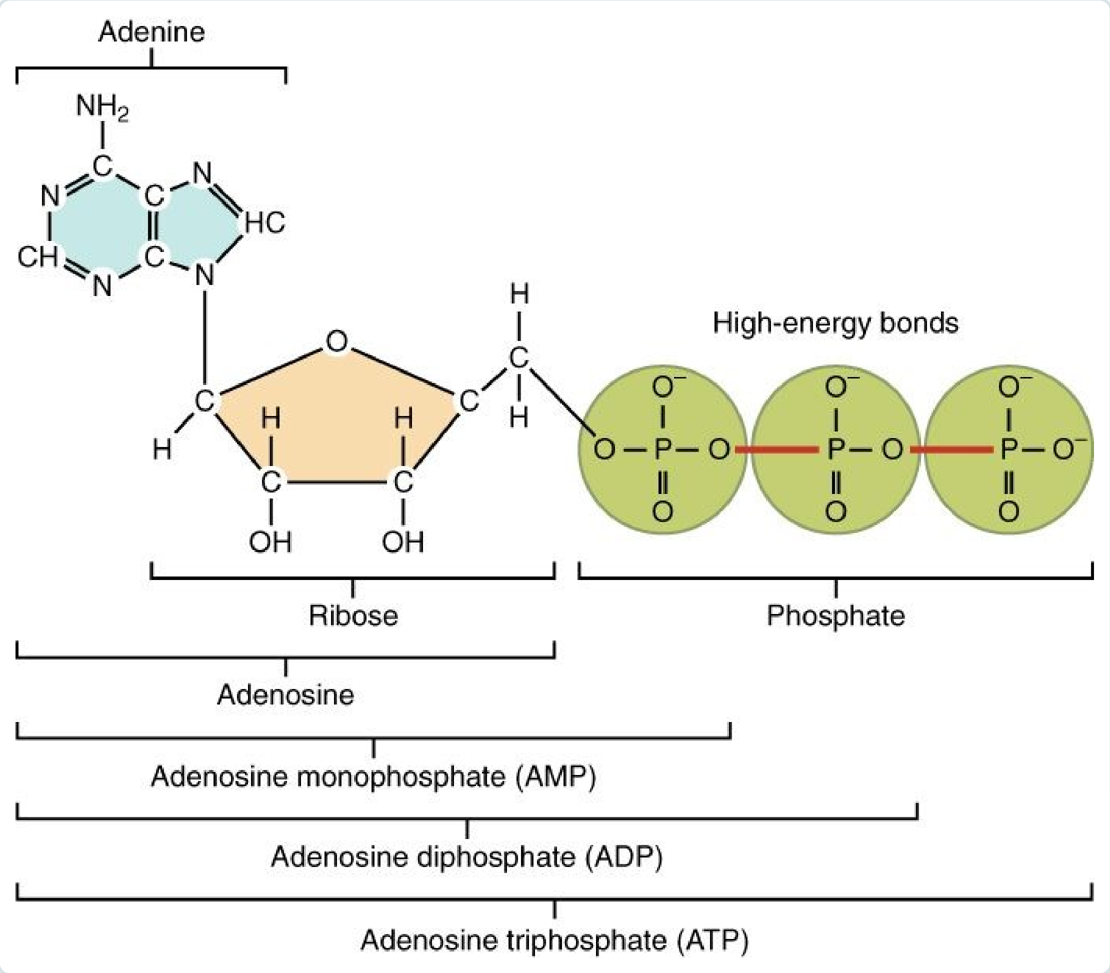
The DNA double helix and base pairing

- DNA and RNA store genetic information
- Built from nucleotides
- Base pairing: A–T and G–C
- The double helix holds the code

SECTION 2.5

ATP — Energy Currency

- Adenosine triphosphate powers the cell
- Energy stored in phosphate bonds
- ATP ⇌ ADP releases and stores energy
- Made mainly in mitochondria



The structure of ATP

CHAPTER 2

Key Takeaways

- All matter is built from elements and atoms; four elements make up most of the body.
- Atoms join through ionic, covalent, and hydrogen bonds.
- Water's polarity makes it the body's solvent and gives it life-supporting properties.
- The body defends a narrow blood pH using buffers, the lungs, and the kidneys.
- Four macromolecules — carbohydrates, lipids, proteins, and nucleic acids — build and run the body.

CLINICAL CONNECTIONS

Why Chemistry Matters in Practical Nursing

Electrolytes

Ions like Na^+, K^+, and Ca^{2+} drive nerve and muscle function — imbalances are dangerous.

Acid–Base Balance

Blood pH must stay near 7.4; acidosis and alkalosis are emergencies.

Blood Glucose

Glucose is the body's fuel — the center of diabetes care.

Nuclear Medicine

Radioisotopes power PET scans and some cancer treatments.

Nutrition

Carbohydrates, fats, and proteins are the macronutrients you counsel patients on.

Enzymes & Fever

High fever can denature enzymes — why dangerous temperatures are treated.